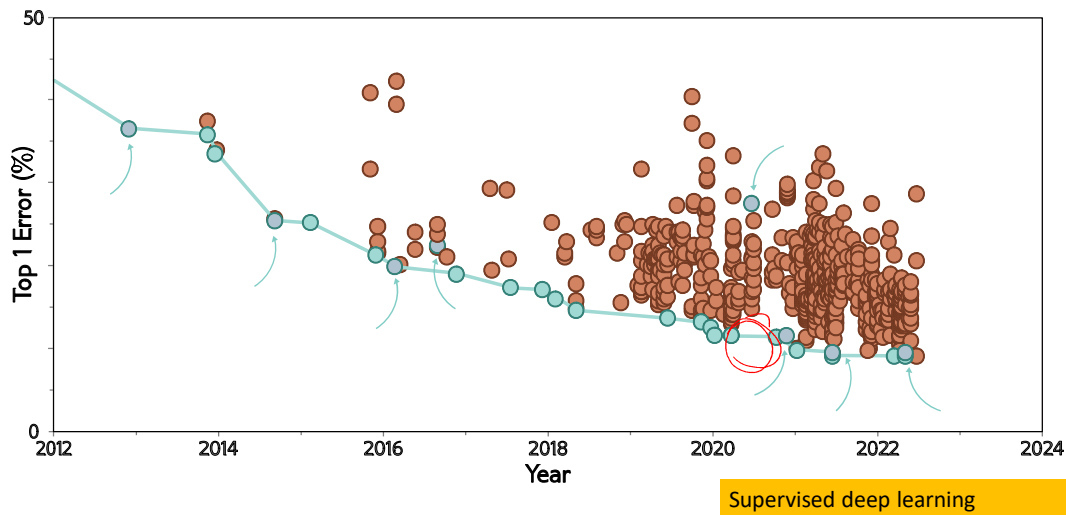
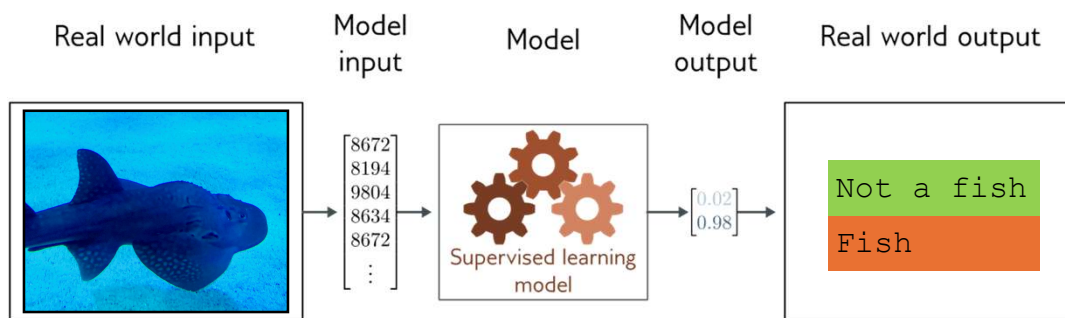


Image Classification Recent History



1

Binary classification



- Goal: predict which of two classes $y \in \{0, 1\}$ the input x belongs to

2

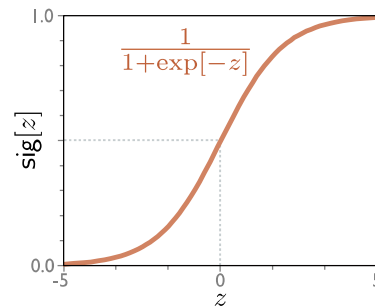
sigmoid

Problem:

- Output of neural network can be anything

Solution:

- Pass through logistic sigmoid function that maps "anything" to [0,1]:



I instance examples ("mini batch")

$$\text{sig}[z] = \frac{1}{1 + \exp[-z]}$$

$$L[\phi] = \sum_{i=1}^I -(1 - y_i) \log [1 - \text{sig}[f[\mathbf{x}_i|\phi]]] - y_i \log [\text{sig}[f[\mathbf{x}_i|\phi]]]$$

minimizing cross entropy

3

Information, Entropy and Cross Entropy

$1 \rightarrow \mathcal{I}(p) = -\log p$
 $\log_2 \frac{1}{2} = \log_2 2^{-1} = -(-1) = 1$
 1-bit inform
 $\frac{0}{4}$

$X \quad X_1 \rightarrow P_1$
 $\quad X_2 \rightarrow P_2$

Entropy $H(X) = -\sum P_i \log p_i$
 $P = (0.9, 0.1) \leftarrow$
 Entropy $-(0.9 \log 0.9 + 0.1 \log 0.1) \approx 0.469$

Suppose $q = (0.6, 0.4)$
 Cross entropy $H(\underline{P}, q) = -(P_1 \log q_1 + P_2 \log q_2)$
 $\underline{\text{true}} \quad \text{model}$
 Example $-(0.9 \log 0.6 + 0.1 \log 0.4) \approx 0.7$
 $0.7 - 0.4 \rightarrow 0.3$

Cross Entropy $\geq$ The Entropy

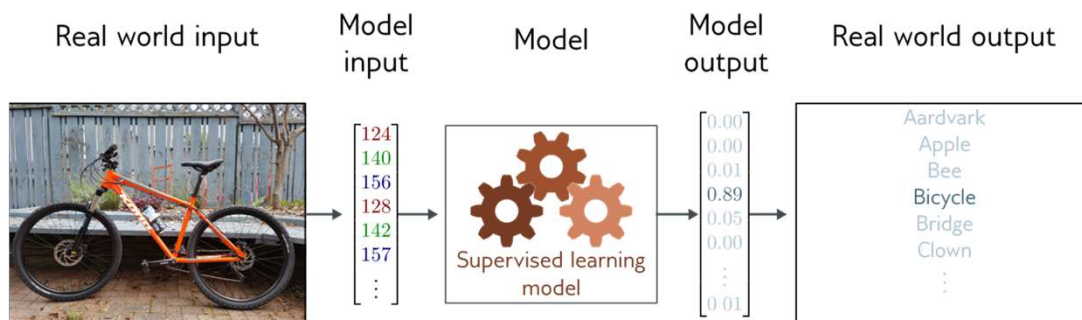
4

Minimizing negative log likelihood

$$\begin{aligned}
 \hat{\phi} &= \operatorname{argmax}_{\phi} \left[\sum_{i=1}^I \log \left[\operatorname{Pr}(\mathbf{y}_i | \mathbf{f}[\mathbf{x}_i, \phi]) \right] \right] \\
 &= \operatorname{argmin}_{\phi} \left[- \sum_{i=1}^I \log \left[\operatorname{Pr}(\mathbf{y}_i | \mathbf{f}[\mathbf{x}_i, \phi]) \right] \right] \\
 &= \operatorname{argmin}_{\phi} \left[L[\phi] \right] \\
 L[\phi] &= \sum_{i=1}^I -(1 - y_i) \log [1 - \operatorname{sig}[\mathbf{f}[\mathbf{x}_i | \phi]]] - y_i \log [\operatorname{sig}[\mathbf{f}[\mathbf{x}_i | \phi]]]
 \end{aligned}$$

5

multiclass classification



Goal: predict which of K classes $y \in \{1, 2, \dots, K\}$ the input x belongs to

6

multiclass classification

$$\mathcal{L}(\hat{\mathbf{y}}, \mathbf{y}) = H(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_{k=1}^K y_k \log \hat{y}_k$$

2. Set the machine learning model $\mathbf{f}[\mathbf{x}, \phi]$ to predict one or more of these parameters so $\boldsymbol{\theta} = \mathbf{f}[\mathbf{x}, \phi]$ and $Pr(\mathbf{y}|\boldsymbol{\theta}) = Pr(\mathbf{y}|\mathbf{f}[\mathbf{x}, \phi])$.

Problem:

- Output of neural network can be anything
- Parameters $\lambda_k \in [0,1]$, sum to one

$$\hat{y}_j = \frac{e^{-z_j}}{\sum_{k=1}^K e^{-z_k}}$$

Solution:

- Pass through function that maps “anything” to $[0,1]$, sum to one

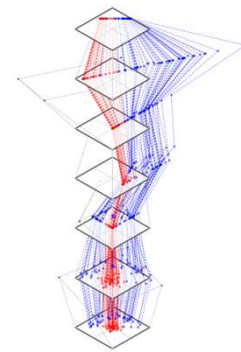
$$\text{softmax}_k[\mathbf{z}] = \frac{\exp[z_k]}{\sum_{k'=1}^K \exp[z_{k'}]}$$

$$Pr(y = k|\mathbf{x}) = \text{softmax}_k[\mathbf{f}[\mathbf{x}, \phi]]$$

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Artificial Neural Nets

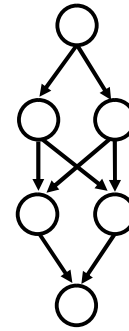
- ANNs are a particular (simple) way of learning: fixing the hypothesis, and a consistent way of optimizing



8

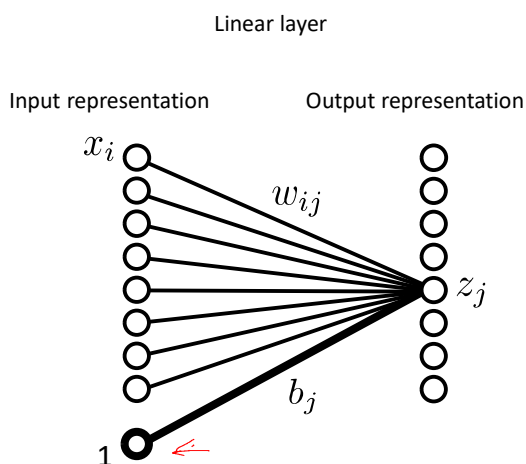
Neural nets

- A directed graph
- Nodes are the neurons. They store data values (inputs, **activations**/hidden representations, outputs). A single neuron stores a scalar.
- Edges store weights. Signals propagate through the edges and get multiplied by the weights.
- A neuron fires, sending a pulse out, when the incoming pulses, from incoming edges, exceed a threshold.



10

Computation in a neural net



$$z_j = [w_{1j} \dots w_{nj}] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + b_j$$

$$z_j = \sum_i w_{ij} x_i + b_j$$

weights

bias

$$z_j = [w_{1j} \dots w_{nj}]$$

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \\ 1 \end{bmatrix}$$

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Shallow Neural Network a)

$$h_1 = a[\theta_{10} + \theta_{11}x]$$

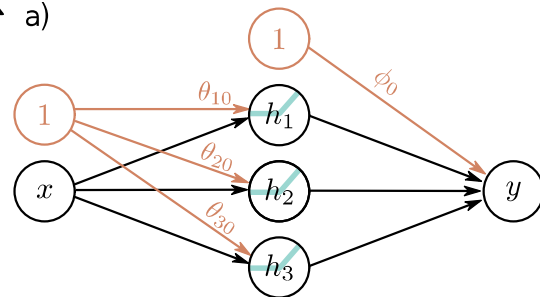
$$h_2 = a[\theta_{20} + \theta_{21}x]$$

$$h_3 = a[\theta_{30} + \theta_{31}x]$$

$$y = \phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3$$

$$y = f[x, \phi]$$

$$= \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x]$$



Each parameter multiplies its source and adds to its target

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Example shallow network

$$y = f[x, \phi]$$

$$= \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x]$$

Activation function

13

Example shallow network

$$y = f[x, \phi]$$

$$= \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x]$$

Activation function

$$a[z] = \text{ReLU}[z] = \begin{cases} 0 & z < 0 \\ z & z \geq 0 \end{cases}.$$

Rectified Linear Unit
(particular kind of activation function)

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Example shallow network

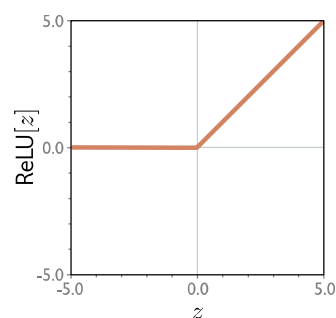
$$y = f[x, \phi]$$

$$= \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x]$$

Activation function

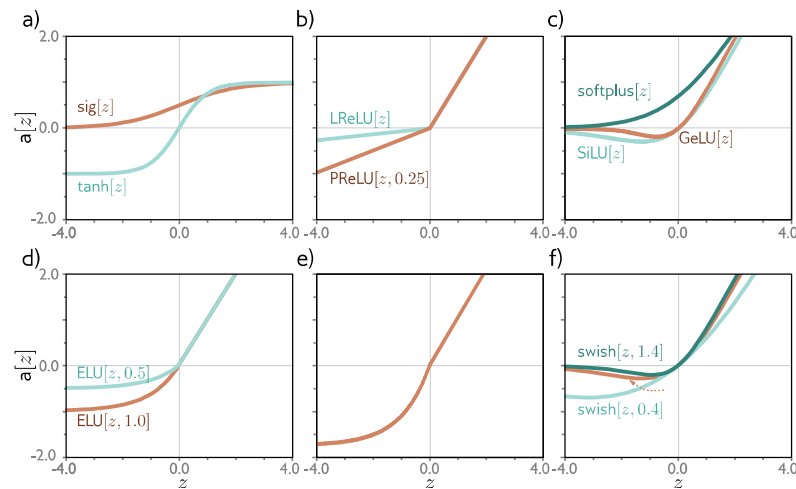
$$a[z] = \text{ReLU}[z] = \begin{cases} 0 & z < 0 \\ z & z \geq 0 \end{cases}.$$

Rectified Linear Unit
(particular kind of activation function)



15

Other activation functions



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Example shallow network

$$y = f[x, \phi]$$

$$= \phi_0 + \phi_1 a[\theta_{10} + \theta_{11}x] + \phi_2 a[\theta_{20} + \theta_{21}x] + \phi_3 a[\theta_{30} + \theta_{31}x]$$

This model has 10 parameters:

$$\phi = \{\phi_0, \phi_1, \phi_2, \phi_3, \theta_{10}, \theta_{11}, \theta_{20}, \theta_{21}, \theta_{30}, \theta_{31}\}$$

- Represents a family of functions
- Parameters determine particular function
- Given parameters can perform inference (run equation)
- Given training dataset $\{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^I$
- Define loss function $L[\phi]$ (least squares)
- Change parameters to minimize loss function

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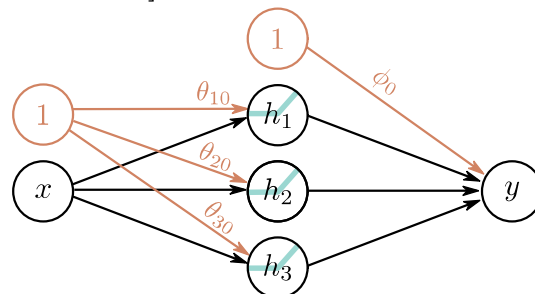
Depicting neural networks

$$h_1 = a[\theta_{10} + \theta_{11}x]$$

$$h_2 = a[\theta_{20} + \theta_{21}x]$$

$$h_3 = a[\theta_{30} + \theta_{31}x]$$

$$y = \phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3$$



Each parameter multiplies its source and adds to its target

18

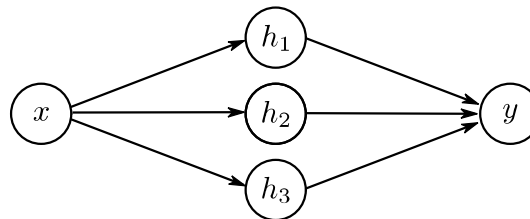
Depicting neural networks

$$h_1 = a[\theta_{10} + \theta_{11}x]$$

$$h_2 = a[\theta_{20} + \theta_{21}x]$$

$$h_3 = a[\theta_{30} + \theta_{31}x]$$

$$y = \phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3$$



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Two outputs

- 1 input, 4 hidden units, 2 outputs

$$h_1 = a[\theta_{10} + \theta_{11}x]$$

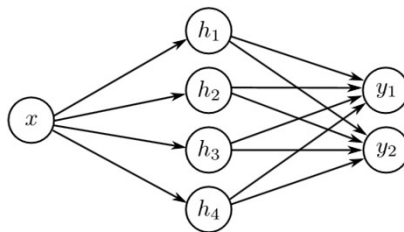
$$h_2 = a[\theta_{20} + \theta_{21}x]$$

$$h_3 = a[\theta_{30} + \theta_{31}x]$$

$$h_4 = a[\theta_{40} + \theta_{41}x]$$

$$y_1 = \phi_{10} + \phi_{11}h_1 + \phi_{12}h_2 + \phi_{13}h_3 + \phi_{14}h_4$$

$$y_2 = \phi_{20} + \phi_{21}h_1 + \phi_{22}h_2 + \phi_{23}h_3 + \phi_{24}h_4$$



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Shallow network with two outputs

- 1 input, 4 hidden units, 2 outputs

$$h_1 = a[\theta_{10} + \theta_{11}x]$$

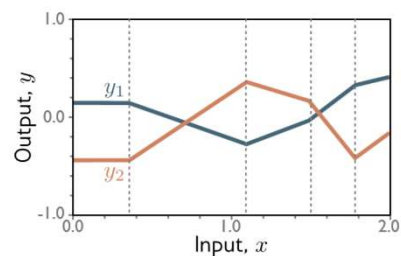
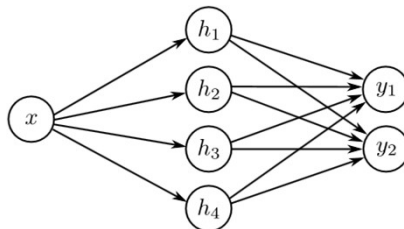
$$h_2 = a[\theta_{20} + \theta_{21}x]$$

$$h_3 = a[\theta_{30} + \theta_{31}x]$$

$$h_4 = a[\theta_{40} + \theta_{41}x]$$

$$y_1 = \phi_{10} + \phi_{11}h_1 + \phi_{12}h_2 + \phi_{13}h_3 + \phi_{14}h_4$$

$$y_2 = \phi_{20} + \phi_{21}h_1 + \phi_{22}h_2 + \phi_{23}h_3 + \phi_{24}h_4$$



21

Two inputs

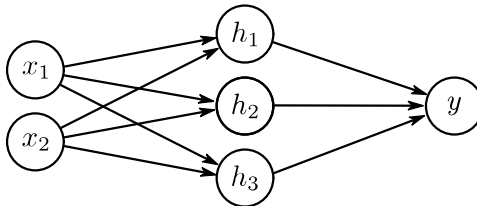
- 2 inputs, 3 hidden units, 1 output

$$h_1 = a[\theta_{10} + \theta_{11}x_1 + \theta_{12}x_2]$$

$$h_2 = a[\theta_{20} + \theta_{21}x_1 + \theta_{22}x_2]$$

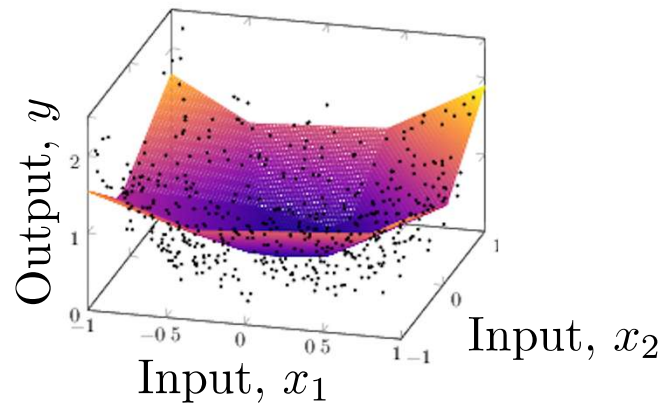
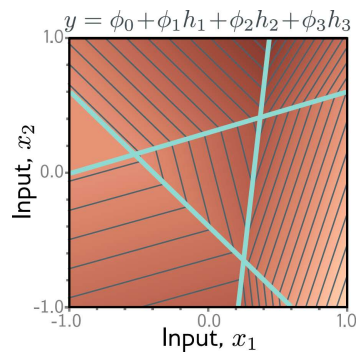
$$h_3 = a[\theta_{30} + \theta_{31}x_1 + \theta_{32}x_2]$$

$$y = \phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3$$



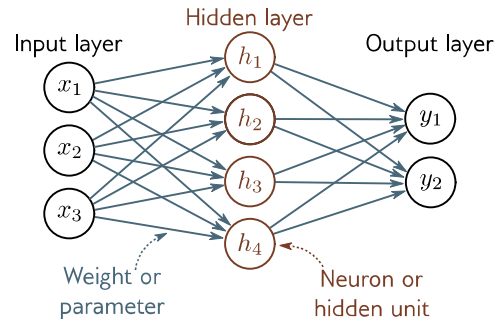
22

Fitting



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Nomenclature



- Everything in one layer connected to everything in the next = **fully connected network**
- No loops = **feedforward network**
- Values after ReLU (activation functions) = **activations**
- Values before ReLU = **pre-activations**
- One hidden layer = **shallow neural network**
- More than one hidden layer = **deep neural network**
- Number of hidden units $\approx$ **capacity**

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Composing two networks.

Network 1:

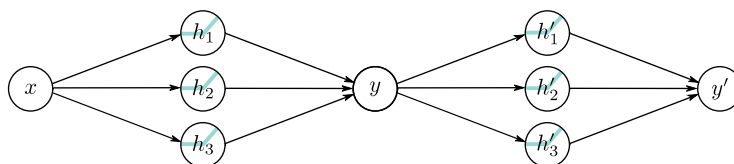
$$\begin{aligned} h_1 &= a[\theta_{10} + \theta_{11}x] \\ h_2 &= a[\theta_{20} + \theta_{21}x] \\ h_3 &= a[\theta_{30} + \theta_{31}x] \end{aligned}$$

$$y = \phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3$$

Network 2:

$$\begin{aligned} h'_1 &= a[\theta'_{10} + \theta'_{11}y] \\ h'_2 &= a[\theta'_{20} + \theta'_{21}y] \\ h'_3 &= a[\theta'_{30} + \theta'_{31}y] \end{aligned}$$

$$y' = \phi'_0 + \phi'_1 h'_1 + \phi'_2 h'_2 + \phi'_3 h'_3$$



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Combine two networks into one

$$\begin{aligned} \text{Network 1:} \quad h_1 &= a[\theta_{10} + \theta_{11}x] \\ h_2 &= a[\theta_{20} + \theta_{21}x] \\ h_3 &= a[\theta_{30} + \theta_{31}x] \end{aligned} \quad y = \phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3$$

$$\begin{aligned} \text{Network 2:} \quad h'_1 &= a[\theta'_{10} + \theta'_{11}y] \\ h'_2 &= a[\theta'_{20} + \theta'_{21}y] \\ h'_3 &= a[\theta'_{30} + \theta'_{31}y] \end{aligned} \quad y' = \phi'_0 + \phi'_1 h'_1 + \phi'_2 h'_2 + \phi'_3 h'_3$$

Hidden units of second network in terms of first:

$$\begin{aligned} h'_1 &= a[\theta'_{10} + \theta'_{11}y] = a[\theta'_{10} + \theta'_{11}\phi_0 + \theta'_{11}\phi_1 h_1 + \theta'_{11}\phi_2 h_2 + \theta'_{11}\phi_3 h_3] \\ h'_2 &= a[\theta'_{20} + \theta'_{21}y] = a[\theta'_{20} + \theta'_{21}\phi_0 + \theta'_{21}\phi_1 h_1 + \theta'_{21}\phi_2 h_2 + \theta'_{21}\phi_3 h_3] \\ h'_3 &= a[\theta'_{30} + \theta'_{31}y] = a[\theta'_{30} + \theta'_{31}\phi_0 + \theta'_{31}\phi_1 h_1 + \theta'_{31}\phi_2 h_2 + \theta'_{31}\phi_3 h_3] \end{aligned}$$

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Create new variables

$$\begin{aligned} h'_1 &= a[\theta'_{10} + \theta'_{11}y] = a[\theta'_{10} + \theta'_{11}\phi_0 + \theta'_{11}\phi_1 h_1 + \theta'_{11}\phi_2 h_2 + \theta'_{11}\phi_3 h_3] \\ h'_2 &= a[\theta'_{20} + \theta'_{21}y] = a[\theta'_{20} + \theta'_{21}\phi_0 + \theta'_{21}\phi_1 h_1 + \theta'_{21}\phi_2 h_2 + \theta'_{21}\phi_3 h_3] \\ h'_3 &= a[\theta'_{30} + \theta'_{31}y] = a[\theta'_{30} + \theta'_{31}\phi_0 + \theta'_{31}\phi_1 h_1 + \theta'_{31}\phi_2 h_2 + \theta'_{31}\phi_3 h_3] \end{aligned}$$

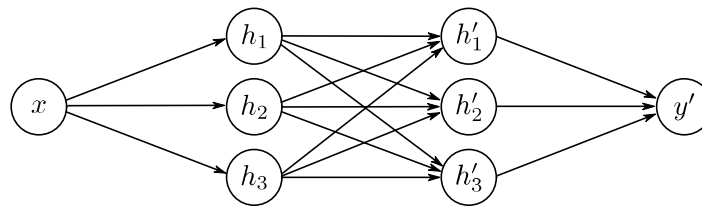
$$\begin{aligned} h'_1 &= a[\psi_{10} + \psi_{11}h_1 + \psi_{12}h_2 + \psi_{13}h_3] \\ h'_2 &= a[\psi_{20} + \psi_{21}h_1 + \psi_{22}h_2 + \psi_{23}h_3] \\ h'_3 &= a[\psi_{30} + \psi_{31}h_1 + \psi_{32}h_2 + \psi_{33}h_3] \end{aligned}$$

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Two-layer network

$$\begin{aligned} h_1 &= a[\theta_{10} + \theta_{11}x] & h'_1 &= a[\psi_{10} + \psi_{11}h_1 + \psi_{12}h_2 + \psi_{13}h_3] \\ h_2 &= a[\theta_{20} + \theta_{21}x] & h'_2 &= a[\psi_{20} + \psi_{21}h_1 + \psi_{22}h_2 + \psi_{23}h_3] \\ h_3 &= a[\theta_{30} + \theta_{31}x] & h'_3 &= a[\psi_{30} + \psi_{31}h_1 + \psi_{32}h_2 + \psi_{33}h_3] \end{aligned}$$

$$y' = \phi'_0 + \phi'_1 h'_1 + \phi'_2 h'_2 + \phi'_3 h'_3$$



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Two-layer network as one equation

$$\begin{aligned} h_1 &= a[\theta_{10} + \theta_{11}x] & h'_1 &= a[\psi_{10} + \psi_{11}h_1 + \psi_{12}h_2 + \psi_{13}h_3] \\ h_2 &= a[\theta_{20} + \theta_{21}x] & h'_2 &= a[\psi_{20} + \psi_{21}h_1 + \psi_{22}h_2 + \psi_{23}h_3] \\ h_3 &= a[\theta_{30} + \theta_{31}x] & h'_3 &= a[\psi_{30} + \psi_{31}h_1 + \psi_{32}h_2 + \psi_{33}h_3] \end{aligned}$$

$$y' = \phi'_0 + \phi'_1 h'_1 + \phi'_2 h'_2 + \phi'_3 h'_3$$

$$\begin{aligned} y' &= \phi'_0 + \phi'_1 a[\psi_{10} + \psi_{11}a[\theta_{10} + \theta_{11}x] + \psi_{12}a[\theta_{20} + \theta_{21}x] + \psi_{13}a[\theta_{30} + \theta_{31}x]] \\ &\quad + \phi'_2 a[\psi_{20} + \psi_{21}a[\theta_{10} + \theta_{11}x] + \psi_{22}a[\theta_{20} + \theta_{21}x] + \psi_{23}a[\theta_{30} + \theta_{31}x]] \\ &\quad + \phi'_3 a[\psi_{30} + \psi_{31}a[\theta_{10} + \theta_{11}x] + \psi_{32}a[\theta_{20} + \theta_{21}x] + \psi_{33}a[\theta_{30} + \theta_{31}x]] \end{aligned}$$

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Notation change #1

$$h_1 = a[\theta_{10} + \theta_{11}x]$$

$$h_2 = a[\theta_{20} + \theta_{21}x]$$

$$h_3 = a[\theta_{30} + \theta_{31}x]$$

$$h'_1 = a[\psi_{10} + \psi_{11}h_1 + \psi_{12}h_2 + \psi_{13}h_3]$$

$$h'_2 = a[\psi_{20} + \psi_{21}h_1 + \psi_{22}h_2 + \psi_{23}h_3]$$

$$h'_3 = a[\psi_{30} + \psi_{31}h_1 + \psi_{32}h_2 + \psi_{33}h_3]$$

$$y' = \phi'_0 + \phi'_1 h'_1 + \phi'_2 h'_2 + \phi'_3 h'_3$$

30

Notation change #1

$$h_1 = a[\theta_{10} + \theta_{11}x]$$

$$h_2 = a[\theta_{20} + \theta_{21}x]$$

$$h_3 = a[\theta_{30} + \theta_{31}x]$$

$$\begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix} = \mathbf{a} \begin{bmatrix} \theta_{10} \\ \theta_{20} \\ \theta_{30} \end{bmatrix} + \begin{bmatrix} \theta_{11} \\ \theta_{21} \\ \theta_{31} \end{bmatrix} x$$

$$h'_1 = a[\psi_{10} + \psi_{11}h_1 + \psi_{12}h_2 + \psi_{13}h_3]$$

$$h'_2 = a[\psi_{20} + \psi_{21}h_1 + \psi_{22}h_2 + \psi_{23}h_3]$$

$$h'_3 = a[\psi_{30} + \psi_{31}h_1 + \psi_{32}h_2 + \psi_{33}h_3]$$

$$y' = \phi'_0 + \phi'_1 h'_1 + \phi'_2 h'_2 + \phi'_3 h'_3$$

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Notation change #1

$$\begin{aligned} h_1 &= a[\theta_{10} + \theta_{11}x] \\ h_2 &= a[\theta_{20} + \theta_{21}x] \\ h_3 &= a[\theta_{30} + \theta_{31}x] \end{aligned} \longrightarrow \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix} = \mathbf{a} \begin{bmatrix} \theta_{10} \\ \theta_{20} \\ \theta_{30} \end{bmatrix} + \begin{bmatrix} \theta_{11} \\ \theta_{21} \\ \theta_{31} \end{bmatrix} x$$

$$\begin{aligned} h'_1 &= a[\psi_{10} + \psi_{11}h_1 + \psi_{12}h_2 + \psi_{13}h_3] \\ h'_2 &= a[\psi_{20} + \psi_{21}h_1 + \psi_{22}h_2 + \psi_{23}h_3] \\ h'_3 &= a[\psi_{30} + \psi_{31}h_1 + \psi_{32}h_2 + \psi_{33}h_3] \end{aligned} \longrightarrow \begin{bmatrix} h'_1 \\ h'_2 \\ h'_3 \end{bmatrix} = \mathbf{a} \begin{bmatrix} \psi_{10} \\ \psi_{20} \\ \psi_{30} \end{bmatrix} + \begin{bmatrix} \psi_{11} & \psi_{12} & \psi_{13} \\ \psi_{21} & \psi_{22} & \psi_{23} \\ \psi_{31} & \psi_{32} & \psi_{33} \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix}$$

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Notation change #1

$$\begin{aligned} h_1 &= a[\theta_{10} + \theta_{11}x] \\ h_2 &= a[\theta_{20} + \theta_{21}x] \\ h_3 &= a[\theta_{30} + \theta_{31}x] \end{aligned} \longrightarrow \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix} = \mathbf{a} \begin{bmatrix} \theta_{10} \\ \theta_{20} \\ \theta_{30} \end{bmatrix} + \begin{bmatrix} \theta_{11} \\ \theta_{21} \\ \theta_{31} \end{bmatrix} x$$

$$\begin{aligned} h'_1 &= a[\psi_{10} + \psi_{11}h_1 + \psi_{12}h_2 + \psi_{13}h_3] \\ h'_2 &= a[\psi_{20} + \psi_{21}h_1 + \psi_{22}h_2 + \psi_{23}h_3] \\ h'_3 &= a[\psi_{30} + \psi_{31}h_1 + \psi_{32}h_2 + \psi_{33}h_3] \end{aligned} \longrightarrow \begin{bmatrix} h'_1 \\ h'_2 \\ h'_3 \end{bmatrix} = \mathbf{a} \begin{bmatrix} \psi_{10} \\ \psi_{20} \\ \psi_{30} \end{bmatrix} + \begin{bmatrix} \psi_{11} & \psi_{12} & \psi_{13} \\ \psi_{21} & \psi_{22} & \psi_{23} \\ \psi_{31} & \psi_{32} & \psi_{33} \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix}$$

$$y' = \phi'_0 + \phi'_1 h'_1 + \phi'_2 h'_2 + \phi'_3 h'_3 \longrightarrow y' = \phi'_0 + \begin{bmatrix} \phi'_1 & \phi'_2 & \phi'_3 \end{bmatrix} \begin{bmatrix} h'_1 \\ h'_2 \\ h'_3 \end{bmatrix}$$

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Notation change #2

$$\begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix} = \mathbf{a} \left[\begin{bmatrix} \theta_{10} \\ \theta_{20} \\ \theta_{30} \end{bmatrix} + \begin{bmatrix} \theta_{11} \\ \theta_{21} \\ \theta_{31} \end{bmatrix} x \right] \longrightarrow \mathbf{h} = \mathbf{a} [\boldsymbol{\theta}_0 + \boldsymbol{\theta}x]$$

$$\begin{bmatrix} h'_1 \\ h'_2 \\ h'_3 \end{bmatrix} = \mathbf{a} \left[\begin{bmatrix} \psi_{10} \\ \psi_{20} \\ \psi_{30} \end{bmatrix} + \begin{bmatrix} \psi_{11} & \psi_{12} & \psi_{13} \\ \psi_{21} & \psi_{22} & \psi_{23} \\ \psi_{31} & \psi_{32} & \psi_{33} \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix} \right] \longrightarrow \mathbf{h}' = \mathbf{a} [\boldsymbol{\psi}_0 + \boldsymbol{\Psi}\mathbf{h}]$$

$$y' = \phi'_0 + \begin{bmatrix} \phi'_1 & \phi'_2 & \phi'_3 \end{bmatrix} \begin{bmatrix} h'_1 \\ h'_2 \\ h'_3 \end{bmatrix} \longrightarrow y = \phi'_0 + \boldsymbol{\phi}'\mathbf{h}'$$

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Notation change #3

$$\mathbf{h} = \mathbf{a} [\boldsymbol{\theta}_0 + \boldsymbol{\theta}x] \longrightarrow \mathbf{h}_1 = \mathbf{a} [\boldsymbol{\beta}_0 + \boldsymbol{\Omega}_0 \mathbf{x}]$$

Bias vector 
Weight matrix 

$$\mathbf{h}' = \mathbf{a} [\boldsymbol{\psi}_0 + \boldsymbol{\Psi}\mathbf{h}] \longrightarrow \mathbf{h}_2 = \mathbf{a} [\boldsymbol{\beta}_1 + \boldsymbol{\Omega}_1 \mathbf{h}_1]$$

$$y = \phi'_0 + \boldsymbol{\phi}'\mathbf{h}' \longrightarrow \mathbf{y} = \boldsymbol{\beta}_2 + \boldsymbol{\Omega}_2 \mathbf{h}_2$$

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General equations for deep network

$$\mathbf{h}_1 = \mathbf{a}[\boldsymbol{\beta}_0 + \boldsymbol{\Omega}_0 \mathbf{x}]$$

$$\mathbf{h}_2 = \mathbf{a}[\boldsymbol{\beta}_1 + \boldsymbol{\Omega}_1 \mathbf{h}_1]$$

$$\mathbf{h}_3 = \mathbf{a}[\boldsymbol{\beta}_2 + \boldsymbol{\Omega}_2 \mathbf{h}_2]$$

$$\vdots$$

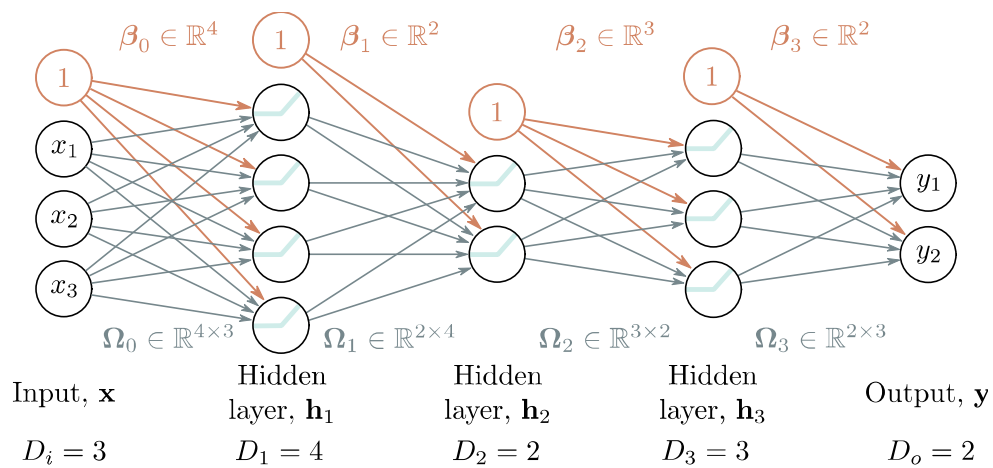
$$\mathbf{h}_K = \mathbf{a}[\boldsymbol{\beta}_{K-1} + \boldsymbol{\Omega}_{K-1} \mathbf{h}_{K-1}]$$

$$\mathbf{y} = \boldsymbol{\beta}_K + \boldsymbol{\Omega}_K \mathbf{h}_K,$$

$$\mathbf{y} = \boldsymbol{\beta}_K + \boldsymbol{\Omega}_K \mathbf{a}[\boldsymbol{\beta}_{K-1} + \boldsymbol{\Omega}_{K-1} \mathbf{a}[\dots \boldsymbol{\beta}_2 + \boldsymbol{\Omega}_2 \mathbf{a}[\boldsymbol{\beta}_1 + \boldsymbol{\Omega}_1 \mathbf{a}[\boldsymbol{\beta}_0 + \boldsymbol{\Omega}_0 \mathbf{x}]] \dots]]$$

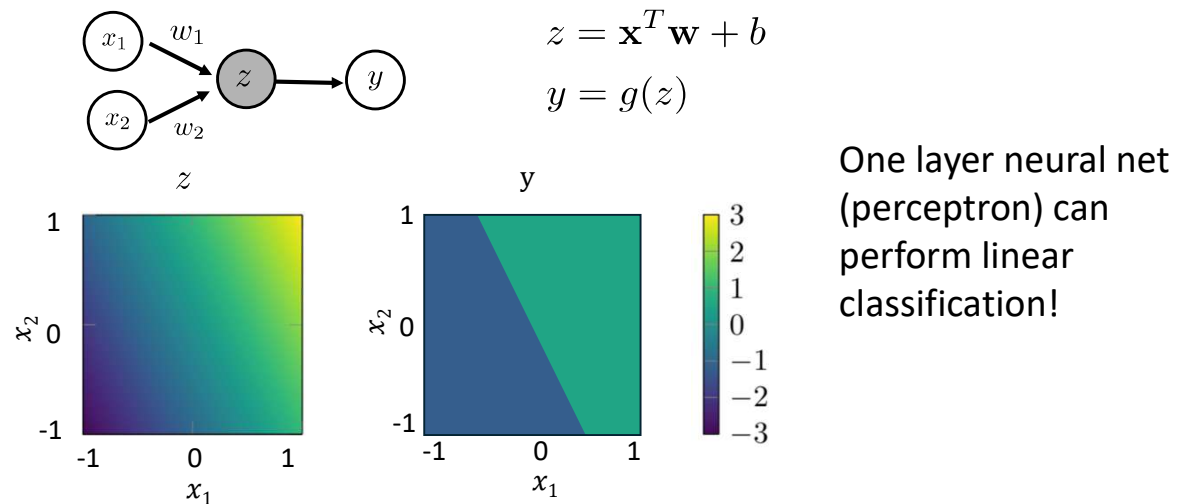
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Example

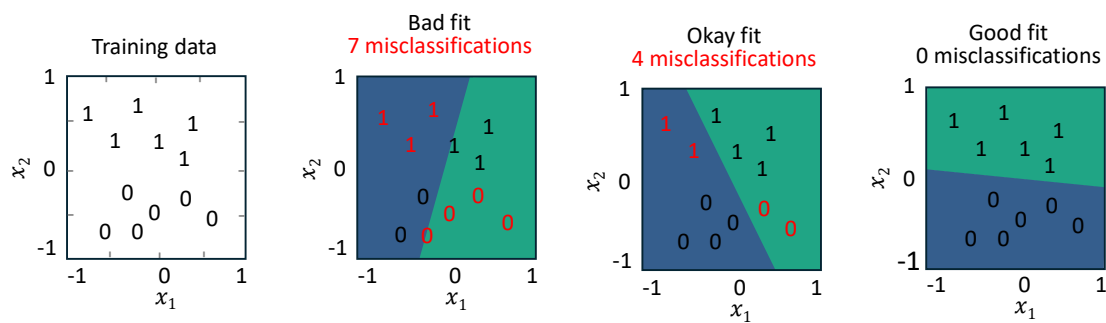


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Example: linear classification with a perceptron



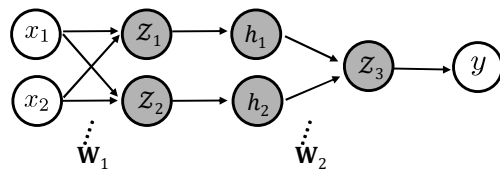
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$$\mathbf{w}^*, b^* = \arg \min_{\mathbf{w}, b} \sum_{i=1}^N \mathcal{L}(g(z^{(i)}), y^{(i)})$$

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Nonlinear classification: Hidden Layer

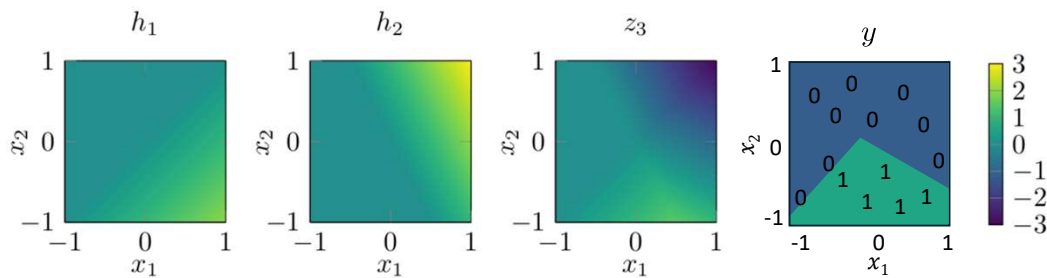


$$\mathbf{z} = \mathbf{W}_1 \mathbf{x} + \mathbf{b}_1$$

$$\mathbf{h} = g(\mathbf{z})$$

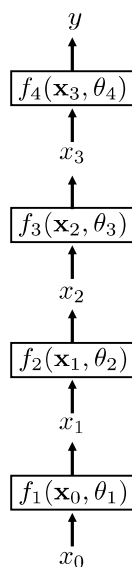
$$z_3 = \mathbf{W}_2 \mathbf{h} + \mathbf{b}_2$$

$$y = 1(z_3 > 0)$$



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Activations vs Parameters

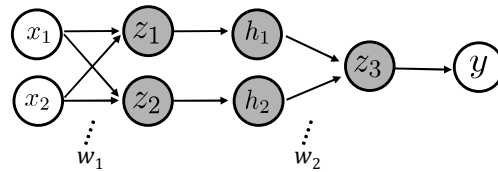


$$[\mathbf{x}, \mathbf{z}_1, \mathbf{h}_1, \dots, \mathbf{z}_{L-1}, \mathbf{h}_{L-1}, \mathbf{y}]$$

- Activations are all values the nodes take on:
- Parameters are the learnable values, θ , mainly the weights and biases
- Typically we optimize the parameters to fit the data, but you can also optimize the activations toward some objective, such as showing the optimal input values to maximize some output decision

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Tensor data structure: Multi-dimensional arrays



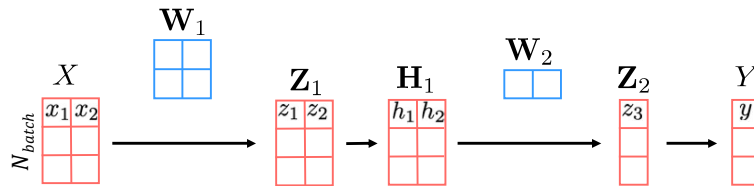
$$\mathbf{z} = \mathbf{W}_1 \mathbf{x} + \mathbf{b}_1$$

$$\mathbf{h} = g(\mathbf{z})$$

$$z_3 = \mathbf{W}_2 \mathbf{h} + \mathbf{b}_2$$

$$y = 1(z_3 > 0)$$

Tensor processing with batch size = 3:



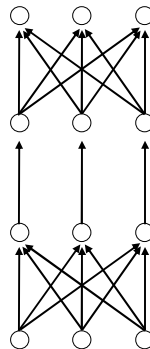
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Multilayer Perceptron

Linear comb of neurons >

Neuron-wise nonlinearity >

Linear comb of neurons >



We can pin down truth either by adding more data, or by using a more constrained architecture.

- + Universal
- + Simple (elegant theory)
- + Embarrassingly parallel
- Weak inductive biases
- Sample inefficient / data hungry
- Dense (fully-connected) linear layers take a lot of compute

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